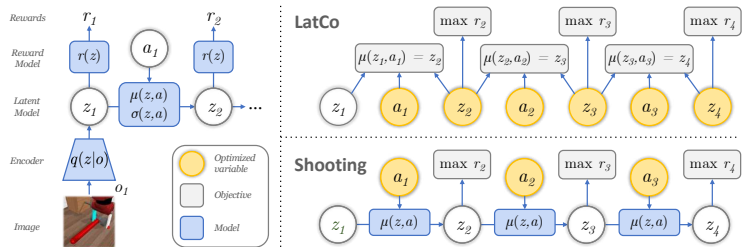


Model-Based Reinforcement Learning via Latent-Space Collocation

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Latent-Space Collocation (LatCo)



Summary:

- Current MBRL methods use simple shooting and fail on long-horizon tasks.
- Collocation is a powerful and principled approach to optimize sequences of states instead of actions, scales better than shooting.
- Use collocation in latent space (LatCo) to scale to visual input.

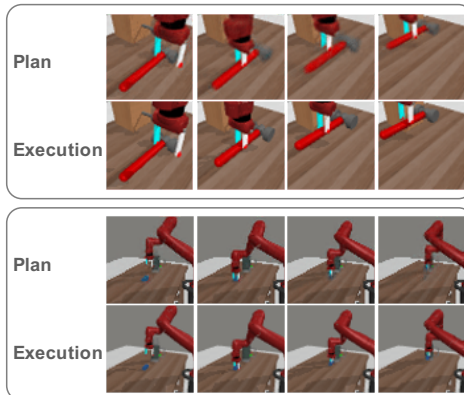
LatCo for Model-Based RL

Algorithm 1 Latent Collocation (LatCo)

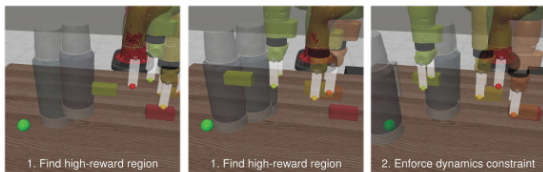
- 1: Start with any available data $\mathcal{D}$
- 2: **while** not converged **do**
- 3: **for** T_{tot} environment steps **with** step T_{cache} **do**
- 4: Infer latent state: $z_t \sim q(z_t | o_t)$
- 5: Define Lagrangian $\mathcal{L}(z_{t+1:t+H}, a_{t:t+H}, \lambda)$:
- 6: Perform K steps of dual gradient descent
- 7: Execute $a_{t:t+T_{\text{cache}}}$ in environment
- 8: Add episode to replay buffer
- 9: **for** It training iterations **do**
- 10: Sample $(o_{1:T}, a_{1:T}, r_{1:T})_{1:b}$ from replay buffer
- 11: Train dynamics model: $\theta := \theta + \alpha \nabla \mathcal{L}_{\text{ELBO}}(o_{1:T}, a_{1:T}, r_{1:T})_{1:b}$

$$\mathcal{L} = \sum_t [r(z_t) - \lambda_t^d (\|z_{t+1} - \mu(z_t, a_t)\|^2 - \epsilon^d) - \lambda_t^a (\max(0, |a_t| - a_m)^2 - \epsilon^a)]$$

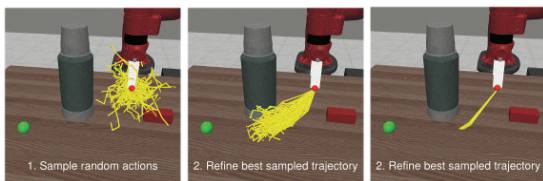
Qualitative Results



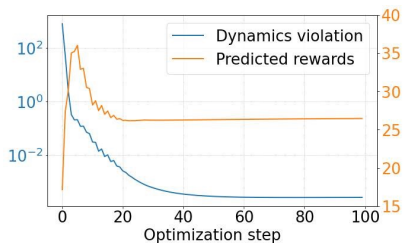
Optimization



LatCo optimization over latent states



CEM optimization over actions



- Levenberg-Marquardt optimizer
- LatCo first converges to high-reward states, and then enforces the dynamics constraint. This initial reward discovery allows it to escape poor local optima.

Quantitative Results

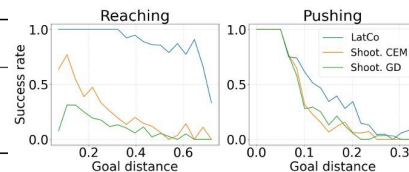
Online Training

	Reaching	Button	Window	Drawer	Pushing	Reacher Easy	Cheetah Run	Quadruped Walk
Shaped reward	×	×	×	×	×	✓	✓	✓
LatCo (Ours)	91 ± 3%	55 ± 4%	49 ± 8%	46 ± 3%	38 ± 3%	559 ± 15	245 ± 12	121 ± 9
PlaNet	20 ± 1%	13 ± 2%	31 ± 2%	22 ± 2%	22 ± 3%	561 ± 14	326 ± 3	72 ± 4
MPPI	16 ± 1%	10 ± 2%	30 ± 2%	21 ± 3%	21 ± 4%	657 ± 17	298 ± 1	57 ± 4
Shoot. GD	8 ± 1%	7 ± 0%	28 ± 1%	18 ± 2%	19 ± 3%	756 ± 7	246 ± 2	101 ± 2

Offline + Online Training

	Thermos	Hammer
Shaped reward	×	×
LatCo (Ours)	37 ± 21%	13 ± 2%
PlaNet	1 ± 1%	1 ± 0%
MPPI	0 ± 0%	0 ± 0%
Shoot. GD	0 ± 0%	3 ± 1%

Task complexity evaluation



Gaussian LatCo

